PRACTICAL 8

Performing Advanced Analytics using Trends, Clustering, and Forecasting

Aim:- To perform advanced analytics in Tableau using trend lines, clustering, distribution analysis, forecasting, and statistical models.

Dataset: Weather & Energy Consumption:

The following dataset is used for performing all the activities in this practical.

Date	Temperature	Humidity	Wind Speed	Energy Consumption
01-01-2025	18	62	8	420
02-01-2025	19	60	7	415
03-01-2025	21	58	6	400
04-01-2025	22	55	5	390
05-01-2025	24	52	5	380
06-01-2025	25	50	4	370
07-01-2025	27	48	4	360
08-01-2025	26	51	5	365
09-01-2025	24	55	6	380
10-01-2025	22	58	7	395
11-01-2025	20	61	8	410
12-01-2025	19	64	9	425
13-01-2025	23	57	6	395
14-01-2025	25	53	5	380
15-01-2025	28	47	4	350

(a) Add and Interpret Trend Lines

Step 1: Connect the Dataset

1. Open Tableau Desktop.
2. Connect to the Excel file containing the Weather & Energy Consumption dataset.
3. Open a new worksheet.
4. Rename the worksheet as Trend Analysis.

Step 2: Create a Scatter Plot

5. Drag Temperature to the Columns shelf.
6. Drag Energy Consumption to the Rows shelf.
7. Tableau creates a scatter plot showing the relationship between temperature and energy consumption.

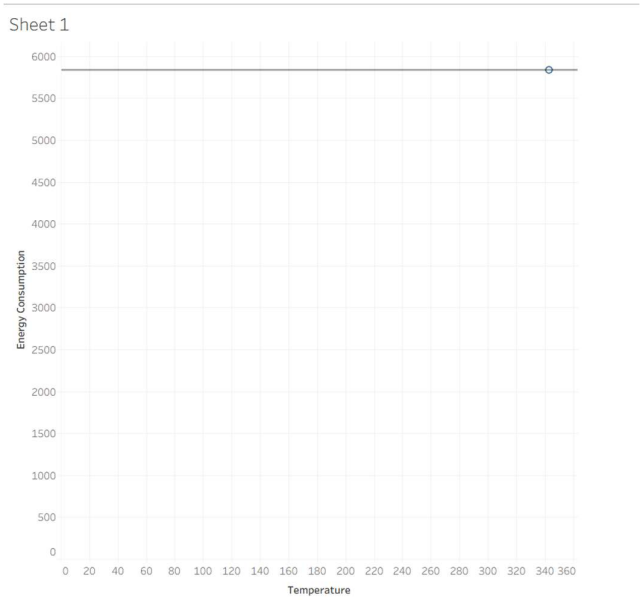
The scatter plot helps us understand whether changes in temperature are related to changes in energy consumption.

Step 3: Add a Linear Trend Line

8. Open the Analytics pane.
9. Locate Trend Line.
10. Drag Trend Line onto the visualization.
11. Select Linear.

A straight trend line is now displayed on the scatter plot.

Expected Output:-



Interpretation:-

The linear trend line represents the overall relationship between Temperature and Energy Consumption.

In this dataset, the trend is generally negative. This means that as temperature increases, energy consumption tends to decrease.

Linear Trend:-

A linear model can be represented as:

$Y = aX + b$

where:

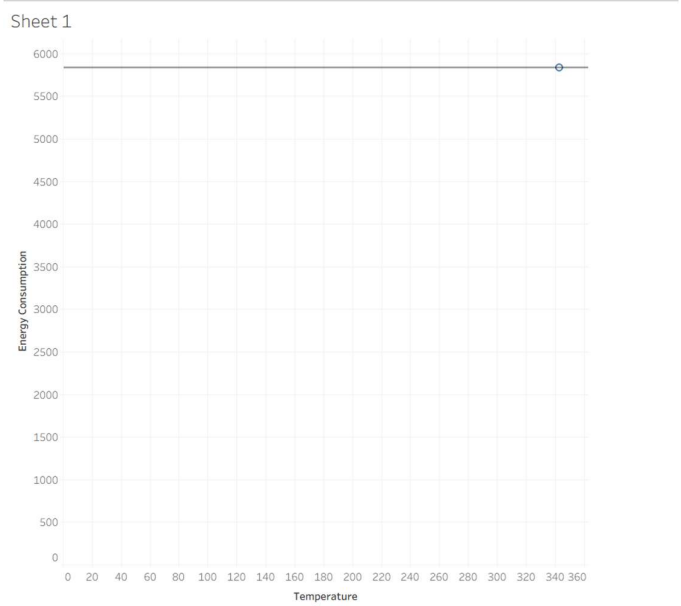
- Y = dependent variable
- X = independent variable
- a = slope
- b = intercept

Step 4: Add an Exponential Trend Line

12. Keep the same scatter plot.
13. Open the Analytics pane.
14. Drag Trend Line onto the visualization.
15. Select Exponential.

Tableau displays an exponential trend line.

Expected Output:-



Exponential Trend:-

An exponential model can be represented as:

$Y = ae^{bx}$

where:

- Y = predicted value
- X = independent variable
- a and b = model parameters

Interpretation of Trend Lines:-

A trend line helps identify the general direction of a relationship between two variables.

- Positive trend: As X increases, Y also increases.
- Negative trend: As X increases, Y decreases.
- Linear trend: Relationship is approximately represented by a straight line.
- Exponential trend: Relationship changes at an increasing or decreasing rate.

(b) Customize and Analyze Trend Models

Step 1: Open Trend Model Details

16. Right-click directly on the trend line.
17. Select Describe Trend Line.

Tableau displays statistical information about the trend model.

The information may include:

- Model equation
- R-squared value
- P-value
- Model type
- Statistical significance
- Model information

Step 2: Understand R-Squared

R-squared (R^2) indicates how well the trend model explains the variation in the data.

The value generally ranges from 0 to 1.

- R^2 close to 1: Very strong model fit.
- R^2 close to 0: Weak model fit.

For example, an R^2 value of 0.90 indicates that the model explains approximately 90% of the variation in the dependent variable.

Step 3: Understand P-Value

The p-value helps determine whether the relationship represented by the model is statistically significant.

A commonly used significance level is:

$p < 0.05$

If the p-value is less than 0.05, the relationship is generally considered statistically significant.

Step 4: Customize the Trend Model

Different trend models can be compared to determine which one provides a better representation of the data.

The main models available in Tableau include:

- Linear
- Logarithmic
- Exponential
- Polynomial
- Power

The appropriate model depends on the pattern present in the dataset.

Interpretation:-

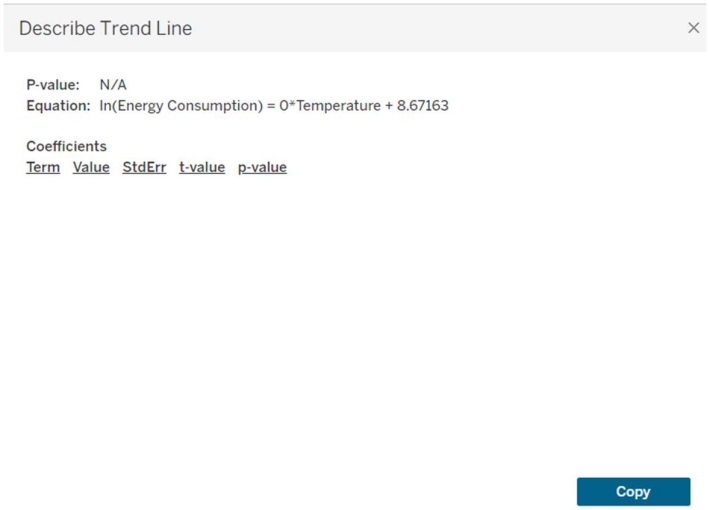
The trend model provides more information than simply looking at the trend line visually.

For example: Linear model

- Shows a straight-line relationship.
- Useful when the rate of change is approximately constant.

Exponential model

- Useful when the rate of change increases or decreases rapidly.
- Can represent relationships where changes occur proportionally.



(c) Perform Clustering on the Dataset

Clustering is used to divide data into groups containing observations with similar characteristics.

Step 1: Create a New Worksheet

18. Create a new worksheet.
19. Rename it as Weather Clustering.

Step 2: Create a Scatter Plot

20. Drag Temperature to Columns.
21. Drag Humidity to Rows.
22. Drag Date to Detail on the Marks card.

Each date is now represented as an individual observation.

Step 3: Apply Clustering

23. Open the Analytics pane.
24. Locate Cluster.
25. Drag Cluster onto the visualization.
26. The Clusters dialog box opens.

Step 4: Select the Number of Clusters

27. Set Number of Clusters to 3.
28. Tableau automatically divides the observations into three groups.

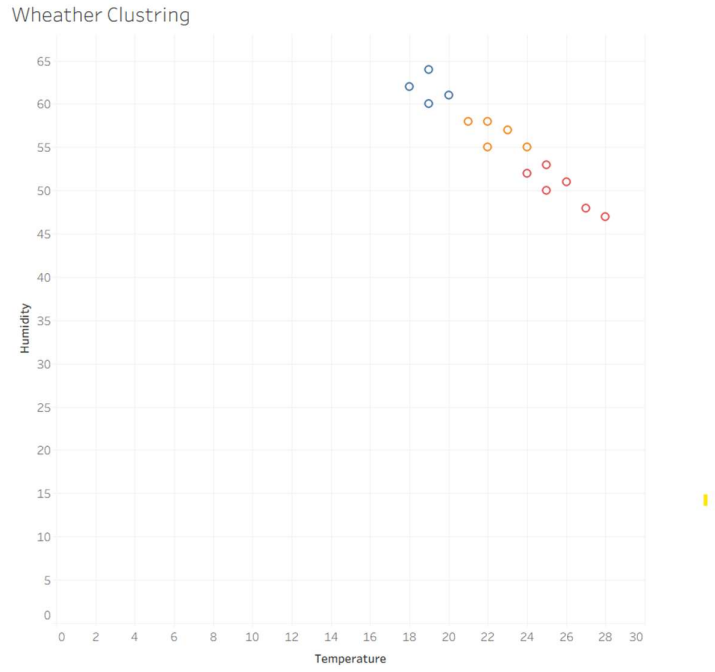
Step 5: Add Variables to the Cluster Model

The following variables can be used for clustering:

- Temperature
- Humidity
- Wind Speed
- Energy Consumption

Add the required fields to the Variables section of the Clusters dialog box.

Excepted Output:-



Interpretation:-

The clusters represent groups of observations having similar characteristics.

For example, the clusters may represent:

- Cluster 1: Higher temperature and lower humidity
- Cluster 2: Moderate temperature and humidity
- Cluster 3: Lower temperature and higher humidity

The exact cluster membership is determined automatically by Tableau.

Conclusion:-

Clustering helps identify natural groups within the dataset without manually assigning categories.

(d) Analyze Distributions Using Built-in Tools

Distribution analysis helps us understand how values are spread across a dataset.

Two useful Tableau tools are:

29. Histogram
30. Box Plot

Part 1: Create a Histogram

Step 1: Create Energy Consumption Bins

31. Right-click Energy Consumption in the Data pane.
32. Select Create → Bins.
33. Set the Size of bins to 20.
34. Click OK.

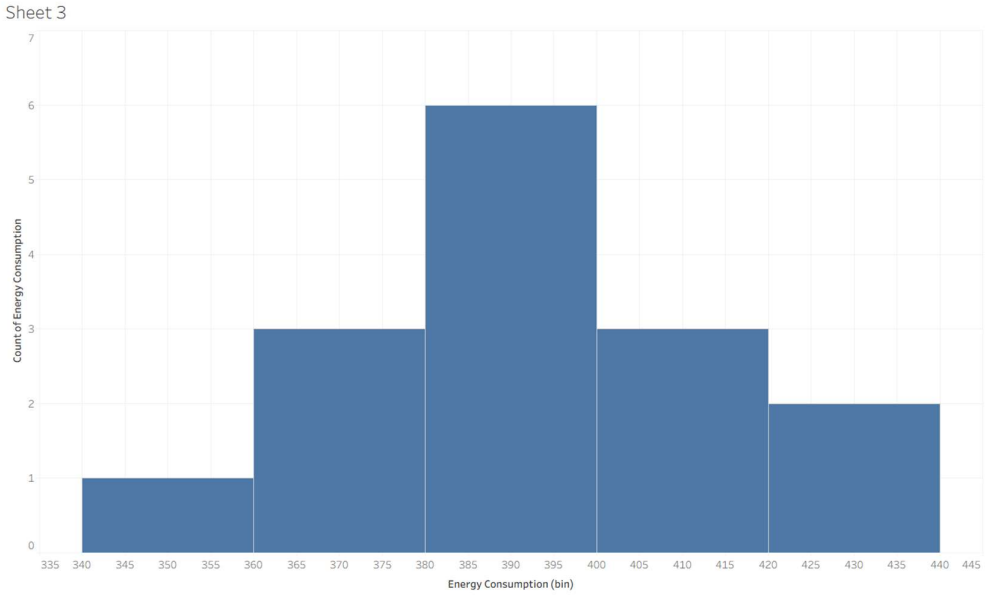
Tableau creates a new field called Energy Consumption (bin).

Step 2: Create the Histogram

35. Drag Energy Consumption (bin) to Columns.
36. Drag Energy Consumption to Rows.
37. If required, change the aggregation to COUNT.

The histogram displays the frequency of energy consumption values within different ranges.

Excepted Output:-



Interpretation of Histogram:-

The histogram helps identify:

- Frequency
- Distribution shape
- Concentration of values
- Possible gaps
- Possible unusual values

For example, if most observations fall between 380 and 420, that range represents the most common energy consumption values in the dataset.

Part 2: Create a Box Plot

Step 1: Create a New Worksheet

38. Create a new worksheet.
39. Rename it as Energy Box Plot.

Step 2: Add Fields

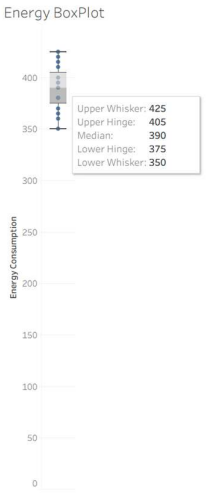
40. Drag Energy Consumption to Rows.
41. Drag Date to Columns.

Step 3: Create the Box Plot

42. Open the Analytics pane.
43. Locate Box Plot.
44. Drag Box Plot onto the visualization.

Alternatively, use Show Me if the Box Plot option is available.

Expected Output:-



Interpretation:-

A box plot displays:

- Median
- Lower quartile
- Upper quartile
- Spread of the data
- Possible outliers

Histogram vs Box Plot:

Histogram	Box Plot
Shows frequency of values	Shows statistical distribution
Uses bins	Uses quartiles
Shows distribution shape	Shows median and spread
Useful for frequency analysis	Useful for identifying outliers

(e) Apply Forecasting Techniques

Forecasting is used to estimate future values based on historical data.

In this practical, Tableau is used to forecast future Energy Consumption.

Step 1: Create a New Worksheet

45. Create a new worksheet.
46. Rename it as Energy Consumption Forecast.

Step 2: Create a Time-Series View

47. Drag Date to Columns.
48. Drag Energy Consumption to Rows.
49. Make sure Date is a true date field.
50. Use a continuous date so that the date appears as a green pill.

Tableau creates a time-series line chart.

Step 3: Apply Forecast

51. Open the Analytics pane.
52. Locate Forecast.
53. Drag Forecast onto the visualization.

Tableau automatically generates a forecast based on the historical energy consumption values.

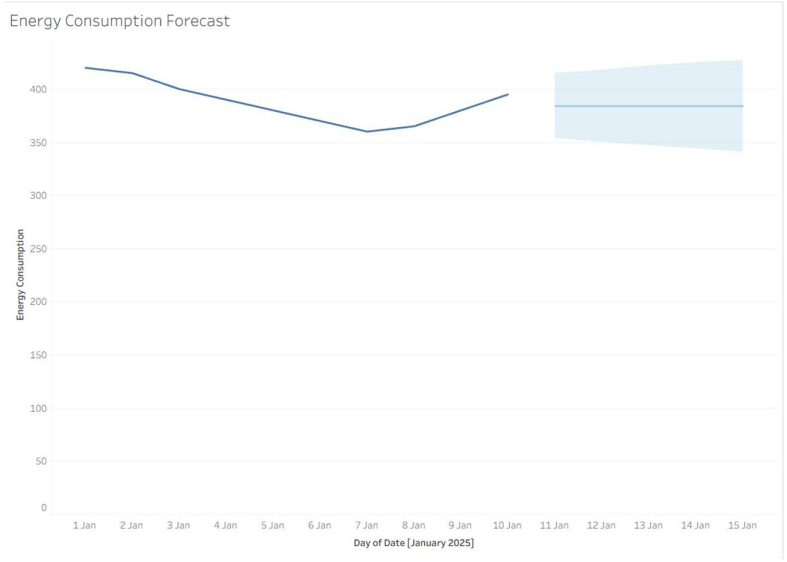
Step 4: Set Forecast Length

To specify the number of future periods:

54. Go to Analysis.
55. Select Forecast.
56. Select Forecast Options.
57. Set the forecast length to the required number of periods.

For demonstration, the forecast length can be set to 5 periods.

Excepted Output:-



Step 5: Analyze the Forecast

The forecast displays:

- Historical values
- Predicted future values
- Forecast range
- Confidence or prediction interval

The forecast line represents the estimated future energy consumption.

What Does Forecasting Do?

Forecasting uses historical patterns to estimate future values.

For example, if the historical data shows that energy consumption is generally decreasing as temperature increases, Tableau uses the available historical pattern to estimate future energy consumption.

The forecast is not an exact prediction. It is an estimate based on the patterns present in the historical data.

Describe the Forecast

To examine the forecasting model:

58. Go to Analysis.
59. Select Forecast.
60. Select Describe Forecast.

Tableau displays information about the forecasting model and its results.

(f) Compare Different Statistical Models

Different statistical models can be compared to determine which model provides a better fit to the data.

In this practical, Linear and Exponential model are compared.

Step 1: Create a Scatter Plot

61. Drag Temperature to Columns.
62. Drag Energy Consumption to Rows.
63. Ensure that the visualization is a scatter plot.

Step 2: Add Linear Trend Line

64. Open the Analytics pane.
65. Drag Trend Line onto the visualization.
66. Select Linear.

Step 3: Analyze the Linear Model

67. Right-click the linear trend line.
68. Select Describe Trend Line.
69. Note the following:

- Equation
- R-squared value

- P-value

Step 4: Add Exponential Trend Line

70. Add another trend line.
71. Select Exponential.
72. Right-click the trend line.
73. Select Describe Trend Line.

Record:

- Equation
- R-squared value
- P-value

Step 5: Compare the Models

Model	Equation	R^2	Interpretation
Linear	$Y = aX + b$	Note Tableau value	Measures straight-line relationship
Exponential	$Y = ae^{bx}$	Note Tableau value	Measures exponential relationship

Interpretation:-

The model with the higher R^2 value generally provides a better fit to the observed data.

However, R^2 should not be considered alone. The model's equation, statistical significance, and the actual pattern of the data should also be considered.

Important Note:

If both Linear and Exponential models show an R^2 value very close to 1, it means that both models fit the current dataset extremely well.

This can happen when the dataset contains a very strong and predictable relationship between the variables.

Therefore, for meaningful model comparison, the dataset should contain some natural variation.

(g) Export and Interpret Statistical Results

Statistical results obtained from Tableau can be recorded and used for further analysis.

Step 1: Open Trend Line Statistics

74. Open the worksheet containing the trend line.
75. Right-click the trend line.
76. Select Describe Trend Line.

Step 2: Examine the Results

The statistical information may include:

- Model equation
- R-squared value
- P-value
- Model type
- Number of observations
- Statistical significance

Step 3: Record the Results

The displayed statistical information can be copied manually into Excel or Word for documentation.

Create a table such as:

Model	**Equation**	**R^2**	**P-value**	**Interpretation**
Linear	Tableau result	Tableau result	Tableau result	Evaluate model fit
Exponential	Tableau result	Tableau result	Tableau result	Evaluate model fit

Step 4: Interpret the Statistical Results

The results can be interpreted using the following guidelines:

R^2

A higher R^2 generally indicates that the model explains more of the variation in the dependent variable.

P-value

A p-value below 0.05 is commonly interpreted as evidence that the relationship is statistically significant.

Equation

The equation describes the mathematical relationship between the independent and dependent variables.

Overall Interpretation:-

Tableau provides several advanced analytical features for understanding data.

Trend lines help identify relationships between variables. Clustering divides observations into groups with similar characteristics. Histograms and box plots help understand the distribution of data. Forecasting estimates future values using historical patterns. Statistical models allow different relationships to be compared using measures such as R^2 and p-value.

These features make Tableau useful not only for creating visualizations but also for performing advanced data analysis and supporting data-driven decision making.

Conclusion:-

In this practical, advanced analytical techniques were performed using Tableau. Linear and exponential trend lines were added and interpreted, statistical models were analyzed using R^2 and p-values, observations were grouped using clustering, distributions were examined using histograms and box plots, and future energy consumption was estimated using forecasting.

Thus, Tableau can be effectively used to perform advanced analytics and interpret relationships, patterns, distributions, and future trends in data.

Main Motive of Practical 8

The main motive of this practical is to learn how to use Tableau not just for making charts, but for performing advanced data analysis.

Students should mainly learn these 5 things:

77. Find relationships → Use Trend Lines to understand whether two variables are related and whether the relationship is positive or negative.
78. Find groups/patterns → Use Clustering to automatically group similar data points.
79. Understand data distribution → Use Histogram and Box Plot to see how data is spread and identify unusual values.
80. Predict future values → Use Forecasting to estimate what may happen in the future based on historical data.
81. Compare statistical models → Use R^2, p-value, and model equations to decide which statistical model fits the data better.

Term	Simple Meaning
Trend Lines	A line that shows the general direction of data, such as whether values are increasing, decreasing, or staying stable.
Clustering	Grouping similar data items together automatically. For example, grouping customers with similar buying habits.
Distribution Analysis	Understanding how data is spread. For example, finding whether most students scored between 60–80 marks.
Forecasting	Using old or past data to predict future values. For example, predicting next month's temperature or future sales.
Statistical Models	Mathematical methods used to analyze data, find patterns, and make predictions.